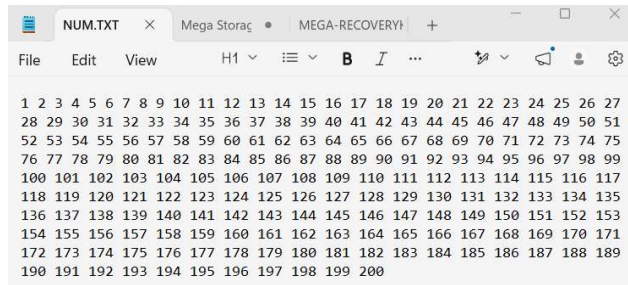
LAB 9

#1

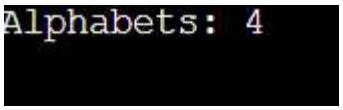
```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("NUM.txt");
    for (int i = 1; i <= 200; i++) {
        fout << i << " ";
    }
    fout.close();
    return 0;
}
```



#2

```
#include <iostream>
#include <fstream>
using namespace std;

void countAlpha() {
    ifstream fin("NOTES.txt");
    char ch;
    int count = 0;
    while (fin.get(ch)) {
        if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' &&
ch <= 'z')) {
            count++;
        }
    }
    cout << "Alphabets: " << count << endl;
    fin.close();
}

int main() {
    countAlpha();
    return 0;
}
```

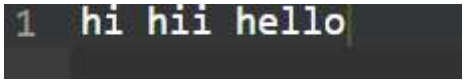
Notes.txt :-



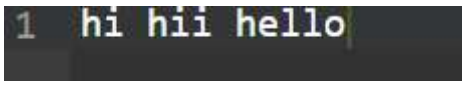
#3

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream fin("NOTES.txt");
    ofstream fout("DEST.TXT");
    char ch;
    while (fin.get(ch)) {
        fout.put(ch);
```

Source



Destination



```
    }
    fin.close();
    fout.close();
    return 0;
}
```

#4

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    char str[100];
    int len = 0;
    cout << "Enter string: ";
    cin.getline(str, 100);

    // length
    while (str[len] != '\0') len++;
    cout << "Length: " << len << endl;

    // write to file
    ofstream fout("data.txt");
    fout << str;
    fout.close();

    // read from file
    ifstream fin("data.txt");
    char ch;
    cout << "From file: ";
    while (fin.get(ch)) {
        cout << ch;
    }

    fin.close();
    return 0;
}
```

```
Enter string: gauri gauri
Length: 11
From file: gauri gauri
```

Data.txt :-

```
1  gauri gauri
```

#5

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    fstream file("AZ.txt", ios::out | ios::in);

    // create A-Z file
    for (char ch = 'A'; ch <= 'Z'; ch++) {
        file.put(ch);
    }

    // (a)
    file.seekg(9);
```

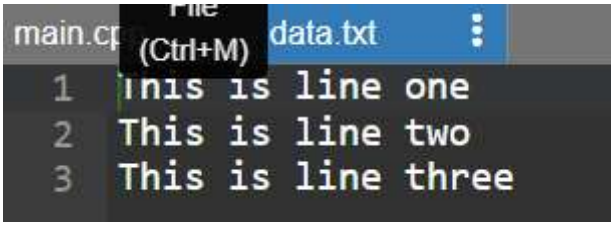
AZ.txt

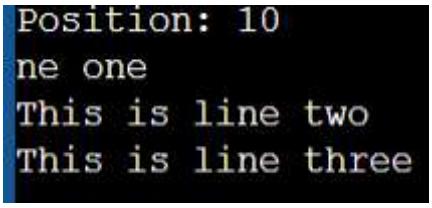
```
ABCDXFGHIJKLMNOPQRSTUVWXYZ
```

```
10th char: J
Size: 26
Last char: Z
```

```
    char ch;
    file.get(ch);
    cout << "10th char: " << ch << endl;

    // (b)
    file.seekp(4);
    file.put('X');

    // (c)
    file.seekg(0, ios::end);
    cout << "Size: " << file.tellg() << endl;

    // (d)
    file.seekg(-1, ios::end);
    file.get(ch);
    cout << "Last char: " << ch << endl;

    file.close();
    return 0;
}
```

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream fout("data.txt");
    fout << "This is line one\nThis is line
two\nThis is line three";
    fout.close();
    ifstream fin("data.txt");

    // (i)
    fin.seekg(10);

    // (ii)
    cout << "Position: " << fin.tellg() << endl;

    // (iii)
    char ch;
    while (fin.get(ch)) {
        cout << ch;
    }
    fin.close();
    return 0;
}
```





#6

```
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    fstream file("hello.txt", ios::out | ios::in);
```

```
    char str[] = "Hello World";
    for (int i = 0; str[i] != '\0'; i++) {
        file.put(str[i]);
        cout << "Position: " << file.tellp() << endl;
    }
    file.seekp(6);
    file << "C++  ";  // spaces to overwrite fully
    file.close();
    return 0;
}
```

```
Position: 9
Position: 10
Position: 11
```